

Mannpreeth Mohankumar Krishnachari
email: mannpreethuk@gmail.com tel: 07387528252

[LinkedIn](#)

Education

MSc in Cybersecurity and Artificial Intelligence - University of Sheffield	2025 – 2026
Bachelor of Technology in Computer Science Engineering – PES University	2023 – 2025
Diploma in Mechatronics – Shree Jayachamarajendra Polytechnic	2017- 2022

Engineering Experience

Research Assistant (MSc Project) – University of Sheffield Dec 2025 – Present

- Working on the RISE: Hardware Roots-of-Trust Reconsidered initiative, developing reconfigurable hardware security primitives for RISC-V architectures.
- Researching data-dependent Physical Unclonable Functions (PUFs), with a focus on DRAM-based PUFs for reconfigurable entropy and secure key generation.
- Designing experiments and analysing PUF quality metrics (entropy, reliability, uniqueness, BER) using Python/MATLAB.
- Collaborating with WIZnet and Siemens Germany GmbH, with outcomes targeting industry adoption and academic publication

Research Assistant – Coacharya Foundation – PES University – Bengaluru May 2025 - August 2025

Researched, designed, and led the development of an AI-enabled digital platform for coaching and client management.

- **Researched solution** requirements and led a cross-functional team in the end-to-end development and evaluation of a **comprehensive digital solution** using React.js, Node.js, and Docker.
- Engineered a dynamic, mobile-friendly website with an integrated **RAG-powered chatbot** (OpenAI, Mistral, Gemini, Llama 3.1) to enhance user engagement and gather user interaction data.
- Developed a data-driven **CRM system** in Python and MySQL that logged session data and translated natural language into SQL queries to support evidence-based decisions by the marketing team.
- Maintained clear documentation and Git/GitHub workflows to ensure reproducibility, traceability, and effective collaboration between the project team, faculty mentors, and industry partners.

Software Developer – Valenta AI – Bengaluru

Nov 2024 – March 2025

- Developed an **AI-powered platform (TensorFlow, PyTorch, Python)** integrated with the **ZOHO Portal** to automate HR data retrieval and processing.
- Contributed to backend logic with **Node.js and REST APIs**, enabling AI-driven analytics that improved data accuracy and decision-making.
- Designed and developed front-end components with **React.js and Next.js**, improving franchisee learning experience.
- Enhanced the platform's **UI/UX** and implemented deployment pipelines using **AWS, Docker, and GitHub Actions**.

PLC Programmer – Siemens – Bengaluru

August 2021 – March 2022

- Designed **fault-tolerant PLC architectures (Siemens TIA Portal, Ladder Logic, C)** that increased system uptime by 60%
- Integrated **PLCs with SCADA systems** to centralize controls and streamline industrial automation processes.
- Applied **cybersecurity best practices** to safeguard automation systems.

Projects at PES University

Uber Clone Web App

- Built a highly scalable ride-hailing platform using **React.js, Node.js**, and **Google Maps API** for real-time location tracking.
- Deployed with **Docker** and versioned in **GitHub**.

Email Forensics Tool

- Created a system in **Python** for tracing email origins and phishing detection using **metadata analysis, NLP techniques, and cybersecurity frameworks**.

Web App Security Assessment

- Conducted **penetration testing** and vulnerability assessments using **OWASP ZAP, Burp Suite, and Python scripting**, identifying critical issues and proposing secure resolutions.

B. Tech Capstone Project - PES University

Crowd Density Estimation and Recommendation System

- Developed a real-time analysis system and interactive web application for predicting crowd density and providing smart recommendations.
- Built the backend using **Flask (Python)** and integrated **YOLO (deep learning, TensorFlow, PyTorch)** for real-time object detection to estimate crowd levels.
- Designed an interactive **dashboard (React.js, Chart.js, HTML/CSS)** for visualizing density trends and recommendations.
- Deployed on **AWS** with **Docker** containers to ensure scalability and seamless user access.

Technical Skills

- **Languages:** Python, Java, JavaScript, Go, C, HTML, CSS, MATLAB
- **AI/ML:** Retrieval Augmented Generation, Large Language Models (OpenAI, Mistral, Gemini), Agents (CrewAI, Autogen), TTS, YOLO, TensorFlow, PyTorch
- **Web Development:** Node.js, React.js, Next.js, REST APIs, Docker, Git
- **Cloud/Database:** AWS, MongoDB, MySQL
- **Design & Simulation Tools:** AutoCAD, SolidWorks, ANSYS (CFD), FEA software, LabVIEW
- **Specialized Areas:** Cybersecurity, Machine Learning, NLP, Robotics, IoT
- **Productivity Tools:** Microsoft Office (Word, Excel, PowerPoint)
- **Soft Skills:** Communication, Teamwork, Problem-Solving, Critical Thinking
- **Embedded Systems:** FPGA, Raspberry Pi, microcontrollers, sensors, actuators, hardware–software co-design and integration

Certificates

- | | |
|----------------------------------|----------|
| • ISC2 – Access Control Concepts | Aug 2025 |
| • ISC2 – Incident Response | Aug 2025 |
| • ISC2 – Security Principles | Aug 2025 |

Hobbies & Interests

Badminton, Cricket, Gym, and Video Games